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AMD DEVELOPER HACKATHON — ACT II · TRACK 3: UNICORN (OPEN INNOVATION)

CarbonPilot

An agentic AI application that analyzes the carbon, energy, and water footprint of AI workloads — and recommends greener GPU and region alternatives, reasoning live on real AMD Instinct compute.

Team: Romir Bulbule & Aarha Khanna **Repo:** github.com/romirbulbule/carbonpilot **Live:** frontend-seven-orpin-28.vercel.app

THE PROBLEM

AI workloads have a hidden environmental cost — and almost nobody can see it

What's missing today

- Teams choose GPUs and regions based on price and availability — carbon impact is rarely part of the decision
- Grid carbon intensity varies 15x between regions (26 gCO2/kWh in Norway vs. 380 in Texas) but almost never factors into deployment choices
- Existing carbon-accounting tools are heavyweight enterprise ESG platforms, not something an ML engineer opens before kicking off a training run

Why it matters now

- AI training and inference workloads are growing faster than grid decarbonization
- GPU choice (AMD Instinct vs. alternatives) and region choice are two of the highest-leverage, lowest-effort levers available — but only if they're visible at decision time
- A tool that reasons about this in plain language, in real time, closes that visibility gap

WHAT WE BUILT

Describe a workload in plain English — get a live, reasoned footprint analysis

CarbonPilot takes a free-text description of an AI workload ("fine-tune a 70B model on 8 MI300X GPUs in Virginia for 24 hours"), and an agent parses it, computes energy/carbon/water impact, and reasons about greener GPU and region alternatives — all visible as a live "Thought → Tool Call → Final Answer" trace, not a black box.

ENERGY USED
144 kWh

CARBON EMISSIONS
64.5 kg CO2e

WATER USAGE
363 liters

GREENER ALT. FOUND
97% savings

HOW IT WORKS

Architecture

Every LLM call and every failure gracefully degrades — Fireworks → Claude → local regex parser — so the app is always demoable, and every response is tagged with which engine actually produced it.

AMD COMPUTE USAGE

✓ Verified against real AMD Instinct hardware

This isn't a claim — here's the real telemetry

We provisioned a real MI300X instance on AMD Developer Cloud, SSH'd in, and ran `rocm-smi` directly. The readings were pushed through CarbonPilot's own live ingest pipeline — the same code path used in the product — and verified readable back out via the API.

```
$ ssh root@[mi300x-instance] "rocm-smi --showpower --showuse --showtemp --json"

{
  "card0": {
    "Temperature (Sensor junction) (C)": "38.0",
    "Current Socket Graphics Package Power (W)": "139.0",
    "GPU use (%)": "0"
  }
}
```

Instance
rocm-7-2-4-gpu-mi300x1-192gb-devcloud-atl1

Agent inference
Fireworks AI (gpt-oss-120b)

Full evidence
docs/amd-hardware-evidence.md in repo

WHAT'S NEXT

Roadmap beyond the hackathon

Carbon-per-token, not carbon-per-GPU-hour

Normalize comparisons by actual compute delivered, not just power draw — a fairer AMD vs. alternative comparison.

Rack/server-level oversight (Tier 2)

Real server power via IPMI/Redfish and PDU metering, for teams running their own hardware rather than cloud instances.

Cloud provider carbon APIs

AWS/GCP/Azure emissions exports as an additional live data source, account-scoped, for teams already on those platforms.

THANK YOU

Try CarbonPilot

GitHub	github.com/romirbulbule/carbonpilot
Live demo	frontend-seven-orpin-28.vercel.app
Team	Romir Bulbule & Aarha Khanna